Social Network Explorer (SNE) – Capstone Report

Title Page

Project Title: Social Network Explorer (SNE)
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Section: A
Subject: Data Structure

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1. Introduction

The Social Network Explorer (SNE) is a command-line based application designed to simulate a simplified social networking system. The project focuses on implementing fundamental data structures and algorithms such as hashing, graphs, breadth-first search (BFS), depth-first search (DFS), and sorting.

This system allows users to create profiles, connect with other users, explore the network, and receive friend recommendations based on shared interests.

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2. Objectives

- To implement user profile management using hashing techniques.
- To represent social connections using graph data structures.
- To compute shortest paths using BFS.
- To explore connections up to a certain depth using DFS.
- To develop a recommendation system based on common interests.
- To design a modular and user-friendly CLI application.

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3. Data Structures Used

3.1 Hash Table (Dictionary)

A dictionary is used to store user profiles. Each user has a unique ID, and their data includes name and interests.

Operations performed:

- Add user

- Get profile
- Update profile

3.2 Graph (Adjacency List)

The social network is represented as a graph using an adjacency list. Each user is a node, and friendships are edges.

Operations performed:

- Add connections
- Remove connections
- Retrieve friends

3.3 Breadth-First Search (BFS)

BFS is used to find the shortest path between two users. It explores the network level by level, ensuring the minimum number of connections.

3.4 Depth-First Search (DFS)

DFS is used to explore the network up to a specific depth. It helps in identifying friends-of-friends and deeper connections.

3.5 Sorting and Hashing (Recommendation System)

The recommendation system suggests new users based on common interests. It uses set intersection to calculate similarity and sorts results based on relevance.

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4. System Design

The project is divided into multiple modules for better organization:

- **profiles.py** – Handles user data
- **graph.py** – Manages connections
- **bfs_dfs.py** – Implements traversal algorithms
- **recommendation.py** – Generates suggestions
- **main.py** – Provides CLI interface
- **test_cases.py** – Contains test cases

This modular design improves readability and maintainability.

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5. Functionalities Implemented

5.1 Profile Management

- Add users
- Update profiles
- Display user information

5.2 Network Management

- Add friendships
- Remove connections
- Display friends list

5.3 Shortest Path (BFS)

- Finds the minimum path between two users
- Displays the path as a list

5.4 Network Exploration (DFS)

- Explores connections up to a given depth
- Helps discover extended networks

5.5 Friend Recommendation

- Suggests users based on mutual interests
- Results are sorted by similarity score

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6. Time Complexity Analysis

Operation	Complexity
Add/Get User	O(1)
Add/Remove Connection	O(1)
BFS Traversal	O(V + E)
DFS Traversal	O(V + E)
Recommendation Sorting	O(n log n)

Where:
V = Number of users
E = Number of connections

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7. Sample Output

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Profiles:
1 : {'name': 'A', 'interests': ['music', 'coding', 'AI']}
2 : {'name': 'B', 'interests': ['sports', 'music']}
3 : {'name': 'C Updated', 'interests': ['coding', 'art']}

Shortest Path:
1 → 6 : [1, 6]

DFS:
Depth 2: [1, 2, 3, 5, 6]

Recommendations:
[(4, 1), (5, 1)]
```

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8. Design Decisions

- Dictionary used for fast access to user profiles.
- Adjacency list chosen for efficient graph representation.
- BFS selected for shortest path due to level-wise traversal.
- DFS used for controlled depth exploration.
- Modular structure used for clarity and scalability.

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9. Challenges Faced

- Managing graph connections correctly.
- Avoiding repeated visits in traversal algorithms.
- Designing an efficient recommendation system.

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10. Conclusion

The Social Network Explorer successfully demonstrates the application of various data structures and algorithms in a real-world scenario. It highlights the importance of graphs in modeling networks and showcases efficient searching and recommendation techniques.

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11. Future Scope

- Development of a graphical user interface (GUI)
- Integration with databases
- Advanced recommendation algorithms
- Location-based friend suggestions
